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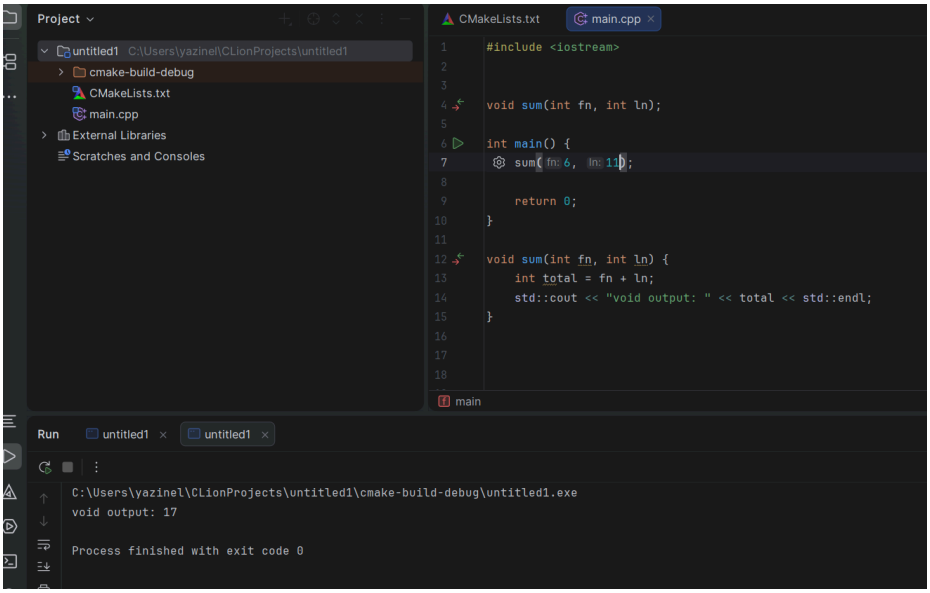
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| | | | |
|---|---|---|---|
| **COURSE CODE/TITLE:** | CPE 010 - Data Structures and Algorithm | **SECTION:** | CPE21S1 |
| **DATE PERFORMED:** | 07/09/26 | **DATE SUBMITTED:** | 7/14/26 |
| **NAME:** | Corro, Yazinel | **INSTRUCTOR:** | Engr.Jim Lord Quejado |

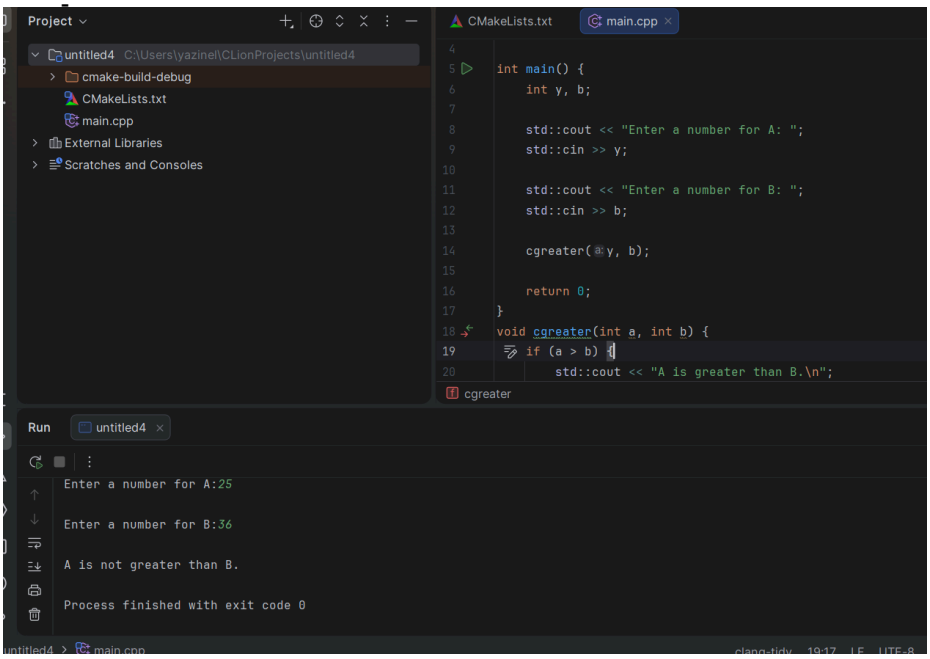
## ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

## 1. PROCEDURE OUTPUT

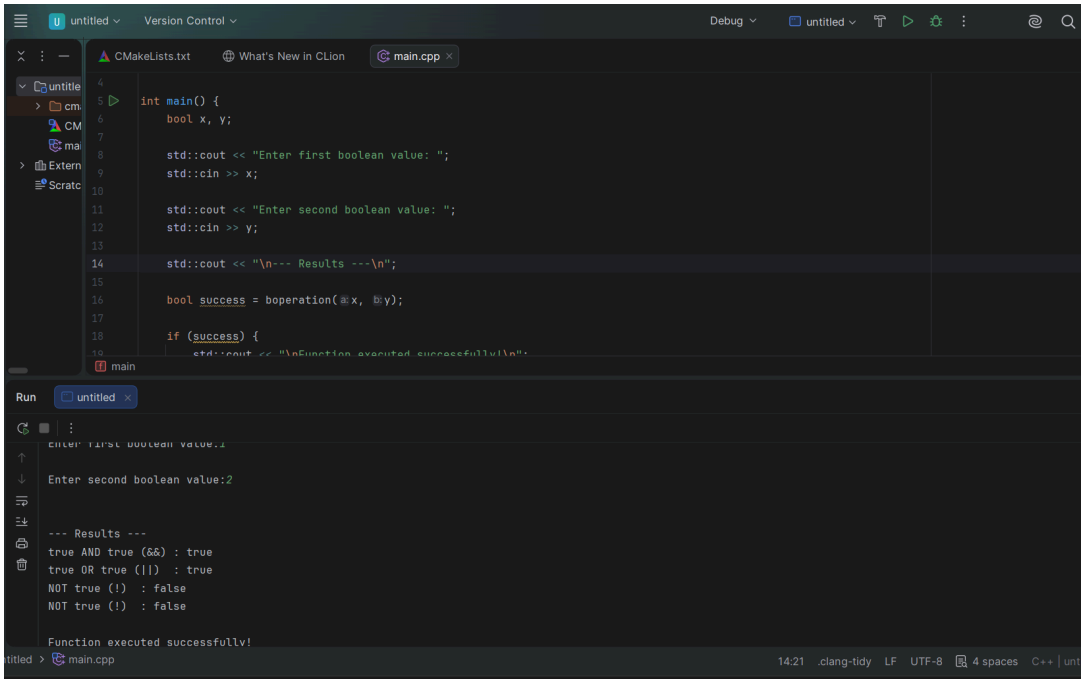
1. Create a function that will take two numbers and display the sum.



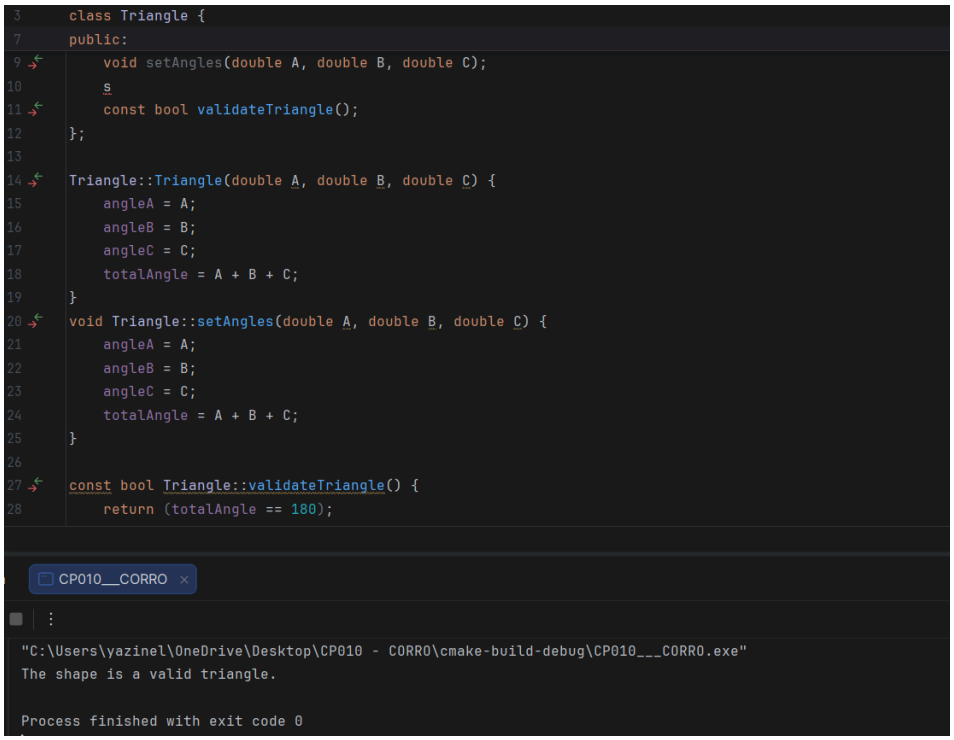

2. Create a function that will return whether variable A is greater than variable B

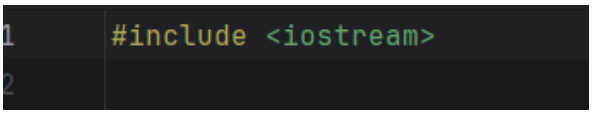
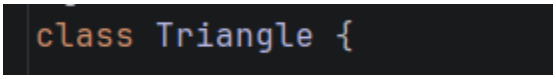
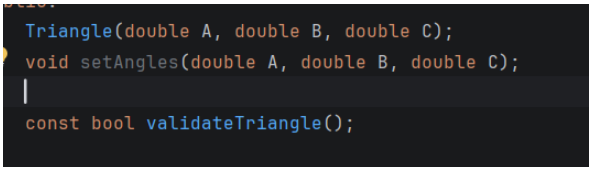
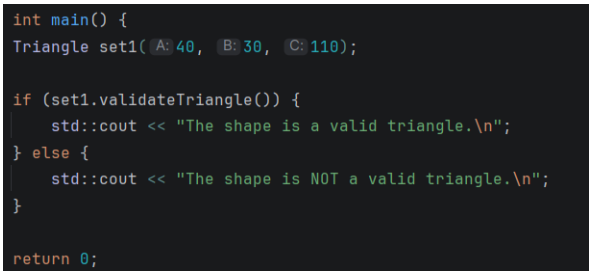
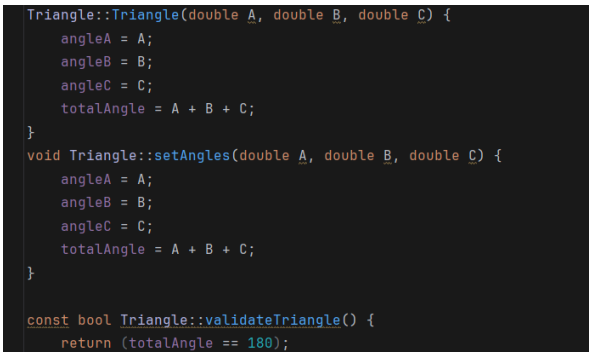


3. Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.



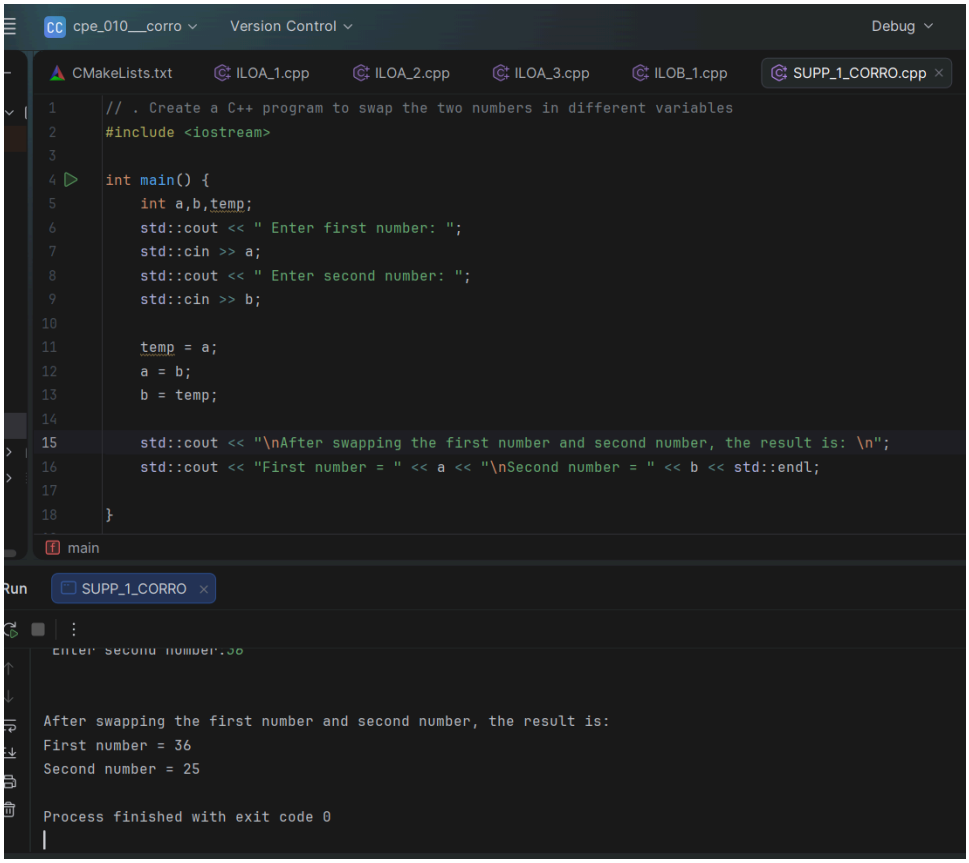

## ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements



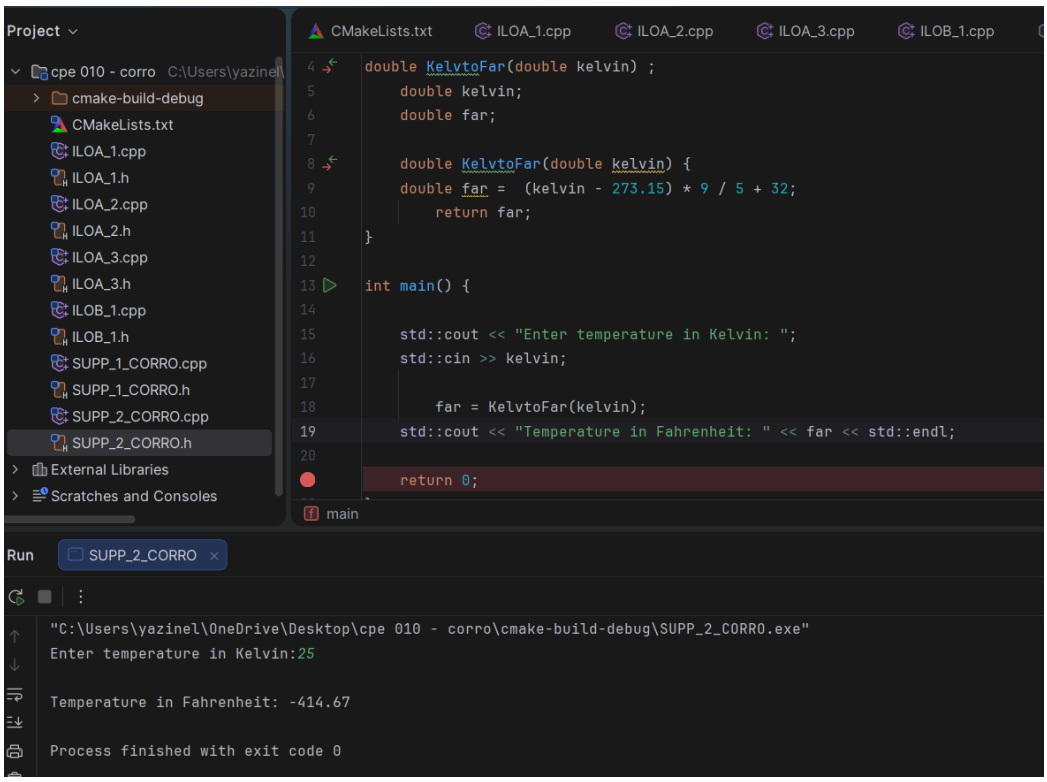
| Sections | Answers |
|---|---|
| Header File Declaration Section |  |
| Global Declaration Section |  |
| Class Declaration and Method Definition Section |  |
| Main Function |  |
| Method Definition |  |

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

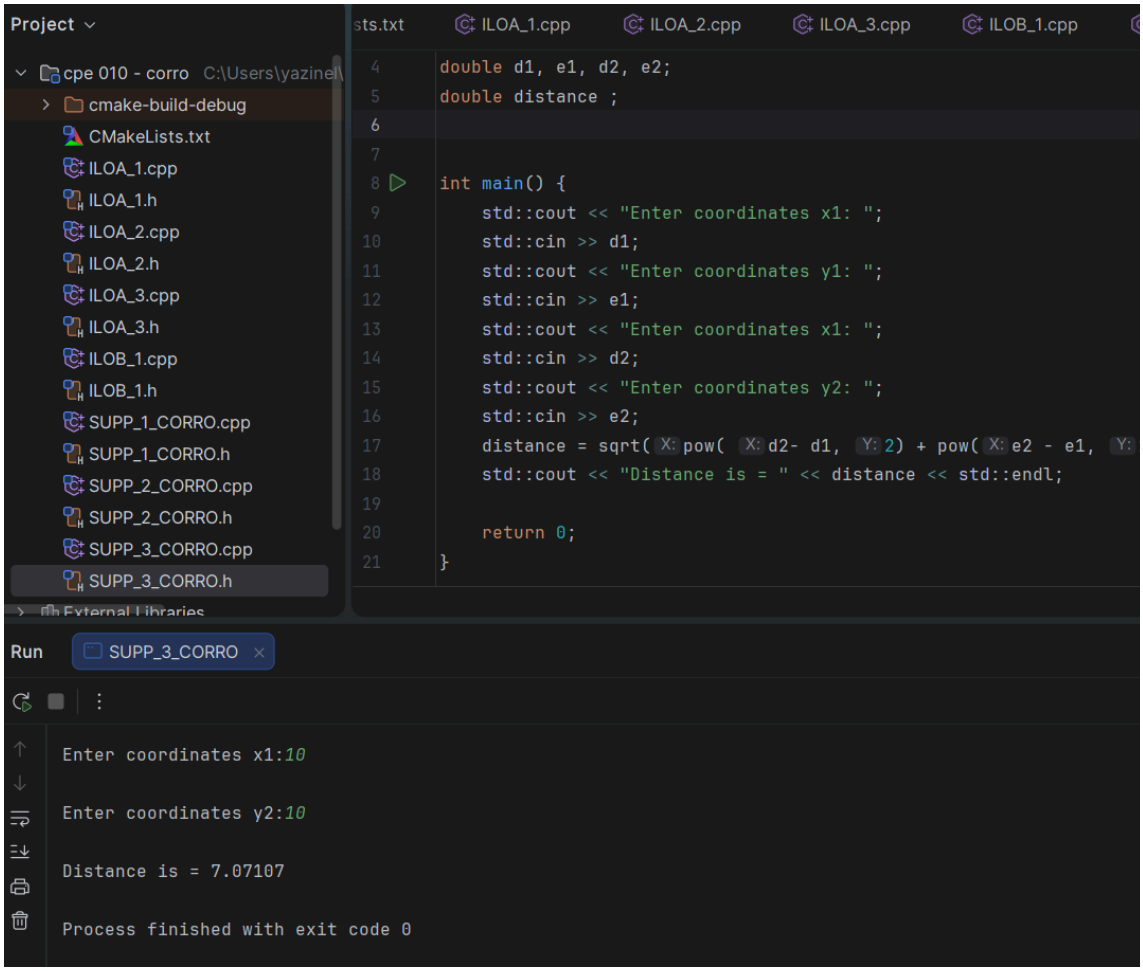
1. . Create a C++ program to swap the two numbers in different variables



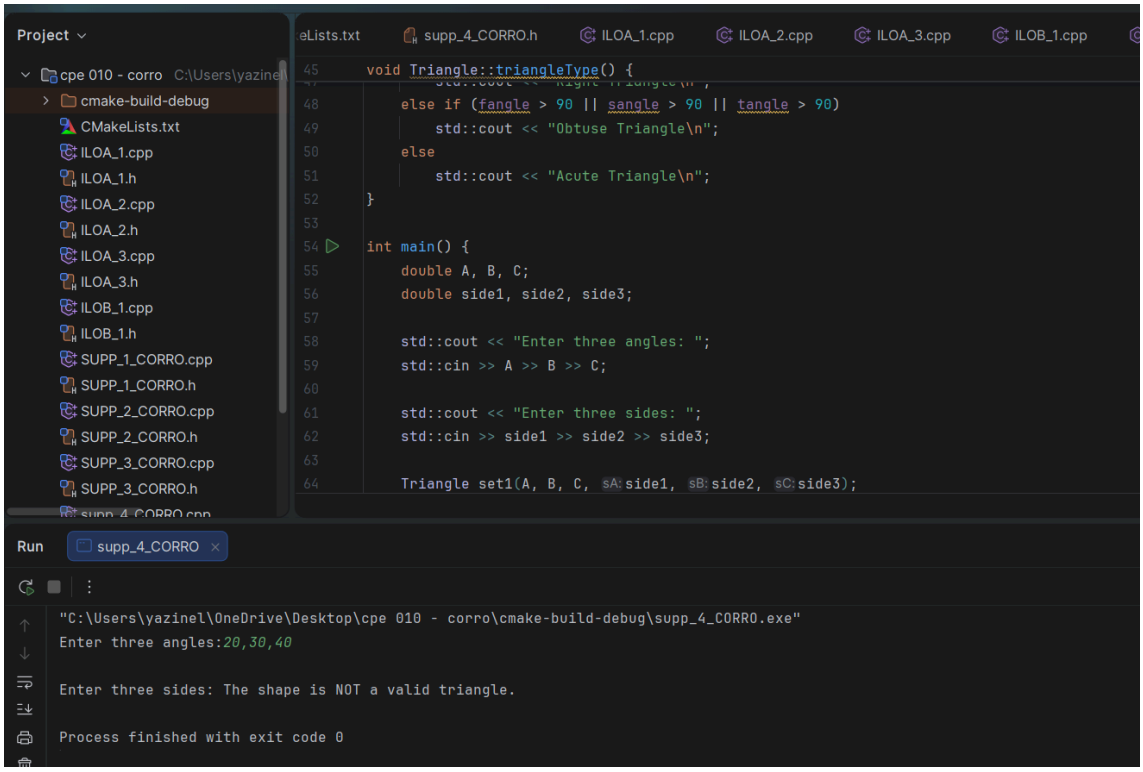

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.



3. Create a C++ program that has a function that will calculate the distance between two points.



4. Modify the code given in ILO B and add the following functions:
   a. A function to compute for the area of a triangle
   b. A function to compute for the perimeter of a triangle
   c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'




| Conclusion: | |
|---|---|
| Summary of lessons learned | Through this activity , I learned how to make and use functions in C++ to do specific jobs like swapping values , changing temperatures , and figuring out the distance between two points . I also learned how to define a class, use constructors, create member function and validate data with a class. |
| Analysis of the procedure | The procedure method teach me how to |

| | |
|---|---|
| | implement C++ concepts step by step. I learned to make functions, use classes and test my program to make sure it worked correctly. I also improved my debugging skills by correcting syntax and compilation errors. |
| Analysis of the supplementary activity | The supplementary activity allowed me to use what I learned in the main activity in different situations. It helped me to improve my problem solving skills and to get more familiar with writing functions and classes. It also taught me how important it is to organize your code correctly. |
| Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement? | I learned a lot of important things about C++ programming in general but I was a little confused about some of the parts, especially fixing errors and working with classes. It was hard but I did get the activity done and through doing it I got to know it better. |

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## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Gemini, Quillbot**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Correction of code and correction of grammar**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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